

Load Dataset

```
url = "https://raw.githubusercontent.com/YBI-Foundation/Dataset/main/Fish.csv"

df = pd.read_csv(url)

df.head()
```

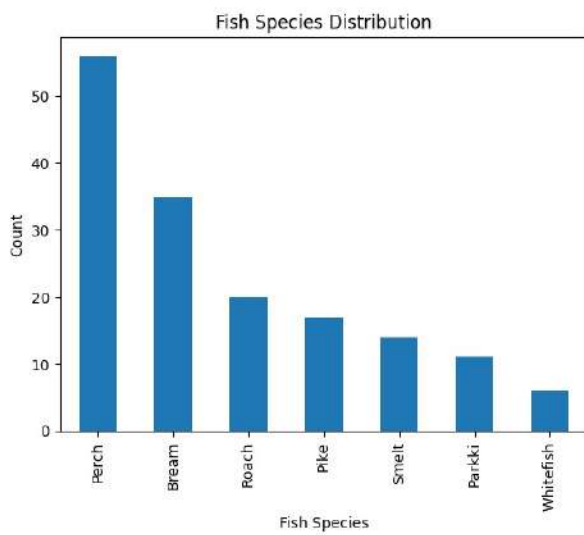
	Category	Species	Weight	Height	Width	Length1	Length2	Length3
0	1	Bream	242.0	11.5200	4.0200	23.2	25.4	30.0
1	1	Bream	290.0	12.4800	4.3056	24.0	26.3	31.2
2	1	Bream	340.0	12.3778	4.6961	23.9	26.5	31.1
3	1	Bream	363.0	12.7300	4.4555	26.3	29.0	33.5
4	1	Bream	430.0	12.4440	5.1340	26.5	29.0	34.0

species count graph

```
df['Species'].value_counts().plot(kind='bar')

plt.xlabel("Fish Species")
plt.ylabel("Count")
plt.title("Fish Species Distribution")

plt.show()
```



Display first five rows

```
print(df.head())
```

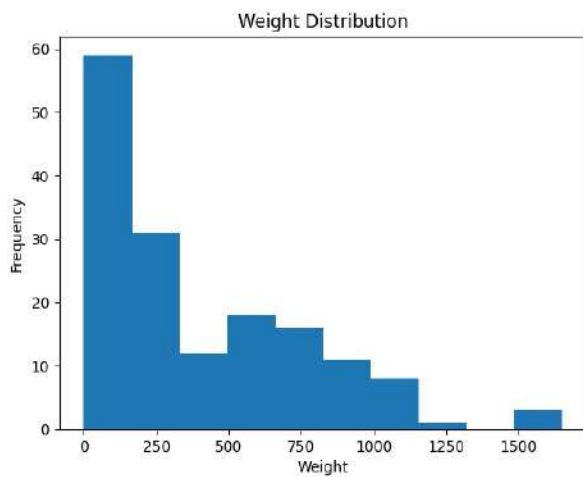
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weight distribution

```
[171] ✓ Out
plt.hist(df['Weight'])

plt.xlabel("Weight")
plt.ylabel("Frequency")
plt.title("Weight Distribution")

plt.show()
```

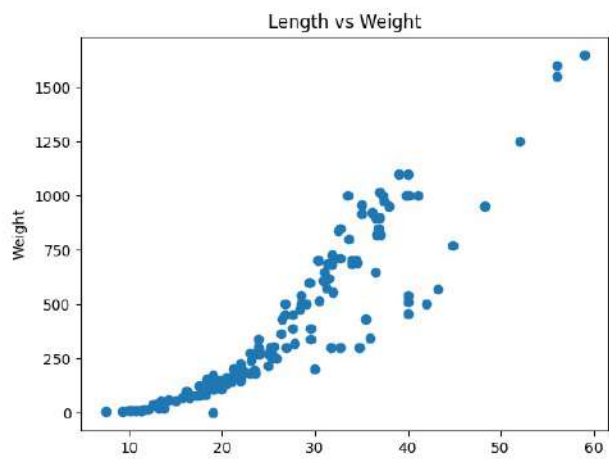


Length vs weight scatter

```
plt.scatter(df['Length1'], df['Weight'])

plt.xlabel("Length")
plt.ylabel("Weight")
plt.title("Length vs Weight")

plt.show()
```



Train Model

```
model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)  
  
model.fit(X_train, y_train)
```

RandomForestClassifier

RandomForestClassifier(random_state=42)

```

from sklearn.metrics import confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns

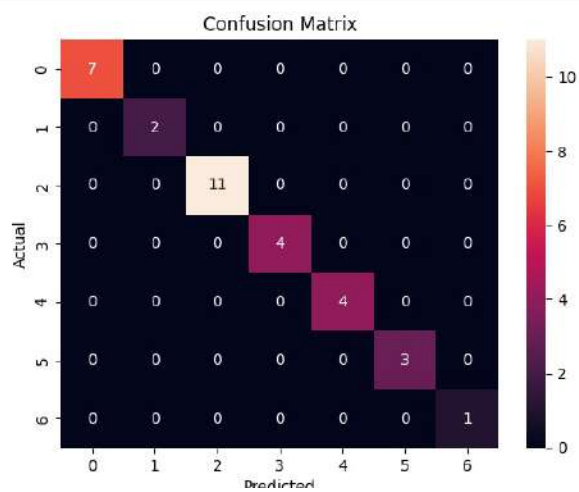
cm = confusion_matrix(y_test,y_pred)

sns.heatmap(
    cm,
    annot=True,
    fmt='d'
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()

```



Accuracy: 1.0

Classification Report

	precision	recall	f1-score	support
Bream	1.00	1.00	1.00	7
Parkki	1.00	1.00	1.00	2
Perch	1.00	1.00	1.00	11
Pike	1.00	1.00	1.00	4
Roach	1.00	1.00	1.00	4
Smelt	1.00	1.00	1.00	3
Whitefish	1.00	1.00	1.00	1
accuracy			1.00	32
macro avg	1.00	1.00	1.00	32
weighted avg	1.00	1.00	1.00	32

Confusion Matrix

[[7 0 0 0 0 0 0]
[0 2 0 0 0 0 0]
[0 0 11 0 0 0 0]
[0 0 0 4 0 0 0]
[0 0 0 0 4 0 0]
[0 0 0 0 0 3 0]
[0 0 0 0 0 0 1]]

✓ 0s

```
plt.bar(  
    X.columns,  
    importance  
)  
  
plt.xlabel("Features")  
plt.ylabel("Importance")  
plt.title("Feature Importance")  
  
plt.xticks(rotation=45)  
  
plt.show()
```



✓

...

